



MUHAMMAD HANAFI

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Electrical Engineering undergraduate at Universitas Muhammadiyah Yogyakarta with strong interests in Instrumentation and Control, Industrial Automation, IoT Systems, Embedded Systems, and Energy Technology. Experienced in industrial internships within Oil & Gas, Power Plant, and IoT-based energy efficiency projects. Skilled in PLC, sensors, microcontrollers, industrial monitoring systems, and automation technologies. Active in robotics research, laboratory assistance, and UAV development projects with hands-on experience in control systems and industrial applications.

Work Experiences

PT PLN Indonesia Power UBP Mrica - Mrica, Banjarnegara,
Central Java, Indonesia

Sep 2025

HAR - Instrument and control

- Assisted in daily preventive and corrective maintenance activities in Instrumentation and Control systems at PLTA UBP Mrica
- Performed inspection, calibration, and troubleshooting of industrial instrumentation devices and control sensors
- Learned and interpreted P&ID diagrams for hydroelectric power plant process systems
- Gained hands-on experience in generator systems, monitoring processes, and power plant operational equipment
- Developed a machine learning-based inflow prediction model to support daily, weekly, and monthly operational planning

PPSDM Migas - Cepu, Central Java, Indonesia

Jan 2025 - Feb 2025

Instrumentation and Control Engineer Intern

- Performed calibration, installation, and troubleshooting of industrial sensors and transmitters used in refinery training systems
- Assisted in analyzing PID control systems and interpreting P&ID diagrams for industrial process applications
- Conducted basic valve maintenance activities including valve disassembly, installation, and inspection procedures
- Operated Allen Bradley PLC systems and learned basic programming, wiring, and industrial control logic implementation
- Observed and practiced basic Distributed Control System (DCS) operations for industrial automation and process control

Telkom Corpu (ITDRI) - Jakarta, Indonesia

Sep 2024 - Dec 2024

IoT Engineer Intern

- Conducted hardware research and testing on imported electronic devices for energy efficiency and IoT implementation projects
- Developed and tested automatic switching and relay systems for controlling high-power electronic equipment, MCBs, and electrical panels
- Monitored and analyzed energy consumption data in Telkom Gatsu Building, a 15-floor office facility
- Learned and implemented IoT communication systems using MQTT protocol for monitoring and automation applications
- Assisted in collecting, documenting, and analyzing electronic equipment data for large-scale IoT deployment planning

Laboratorium Teknik Elektro Universitas Muhammadiyah
Yogyakarta - Yogyakarta, Indonesia

Oct 2023 - Jun 2026

Laboratory Assistant

- Assisted more than 100+ students in conducting laboratory experiments in Basic Programming, Physics, Electronic Circuits, Robotics, Microprocessor Systems, and Industrial Automation
- Guided students in programming activities using C++, Python, and microcontroller-based system development
- Supported industrial automation practicum activities using Omron PLC systems and control logic implementation
- Assisted students in troubleshooting electronic circuits, sensors, embedded systems, and automation hardware
- Evaluated and graded practicum reports based on technical analysis, programming accuracy, and system performance
- Coordinated with laboratory staff and fellow assistants to ensure smooth laboratory operations and practicum activities

Education Level

Universitas Muhammadiyah Yogyakarta - Yogyakarta, Indonesia

Sep 2022 - Jun 2026

Bachelor of Electrical Engineering, 3.85/4.00

- Relevant Coursework: Instrumentation and Control Systems, Industrial Automation, Microprocessor Systems, Embedded Systems and IoT, Electronic Circuits, Basic Programming and Robotics, PLC and Industrial Control, Sensors and Industrial Instrumentation, Machine Learning Fundamentals, Electrical Power Systems

Organisational Experience

Microcontrollers and Robotics Club (MRC) - Yogyakarta, Indonesia

Jan 2023 - Oct 2024

Head of VTOL Division

- Led VTOL UAV research and development activities focusing on autonomous control and flight system integration

- Coordinated a team of 10 members in UAV system development, testing, and competition preparation
- Developed technical workflows and evaluation systems to improve team productivity and project efficiency
- Assisted in workshop and technical training activities related to robotics, UAV systems, and embedded programming
- Represented the university as a Finalist in the Indonesian Flying Robot Contest (KRTI) organized by the Ministry of Education, Culture, Research and Technology of the Republic of Indonesia

Keluarga Mahasiswa Teknik Elektro (KMTE) - Yogyakarta, Indonesia

Aug 2024 - Aug 2025

Organization Staff

- Assisted in organizing academic and student development activities within the Electrical Engineering department
- Coordinated with internal divisions and committees to support organizational programs and events
- Supported communication and collaboration between students, committees, and organization members
- Assisted in managing event preparation, documentation, and operational activities
- Contributed to teamwork and organizational development activities within KMTE

ICOMMIT UMY - Yogyakarta, Indonesia

Nov 2023 - Nov 2025

Research and Technology Staff

- Assisted in research and development activities related to IoT systems, embedded systems, and smart technology applications
- Developed and tested microcontroller-based systems using sensors, communication modules, and automation devices
- Supported IoT communication implementation using MQTT protocol and wireless communication systems
- Collaborated with research teams in developing monitoring and automation projects based on IoT technology
- Assisted in technical discussions, project documentation, and technology development activities within the organization

Certifications

- **Belajar Elektro - Introduction to Electrical Instrumentation**  (2025)
- **Belajar Elektro - Understanding Electrical Instrumentation and Control in Manufacture**  (2025)
- **Belajar Elektro - Understanding Electrical Instrumentation and Control in Oil and Gas Industry**  (2025)
- **BNSP - Teknisi Senior Otomasi Industri**  (2025)
- **Kelas Engineer - SCADA AUTOMATION ELECTRICAL BOOTCAMP** (2026)
- **ICS Cademy - Paket Belajar PLC 7in1 dan Scada AVEVA** (2026)